



The Open University

Mathematics, Computing and Technology: Level 3
M362 Developing concurrent distributed systems

M362

Glossary

A

abort Deliberately stop something that has started.

Aborted (transaction state) The state a transaction is in after it has been rolled back to undo any partially completed actions of the transaction.

access controller A component used by the security manager to check permissions belonging to classes.

ACID properties The desirable properties of a transaction: atomicity, consistency, isolation and durability.

acquire phase The phase in two-phase locking in which the locks are gradually acquired, and in which no locks can be released.

action (in a policy file) In Java, the part of an entry in a policy file that describes the kinds of action that may be performed with respect to some target.

action-oriented model A software model in which the actions carried out by processors are central, as opposed to an object-oriented model where objects are central, or a **data-oriented model** where data is central.

activation frames A data structure storing information about a method call in Java that allows the method to resume after being placed on a stack.

Active (transaction state) The state a transaction is in as soon as it has started, and in which it remains while executing.

addressing Identifying a particular computer on the internet using either a symbolic address or a numeric address.

ad-hoc network A network with dynamic routing and without dedicated nodes like routers and hubs, thereby allowing nodes to easily join and leave the network.

after image In logging, the name given to the value of a variable after an update is made.

annotation A Java language construct that provides additional information about the class, method, variable, etc. that is being annotated. The information can be used at compile time or at runtime, typically to generate code, enforce additional constraints, or provide extra documentation. Each annotation has zero or more elements, each being a name-value pair.

applet A block of compiled Java (in bytecode form) that can be downloaded along with a web page and that runs within suitable browsers.

application (Java) A Java program that can run independently of a browser.

application client A Java EE client implemented as a Java application, running on the client computer either in its own JVM or in an application client container.

application client container A software environment for application clients that takes care of many of the routine requirements of communication, security and so on. In particular, it facilitates access from an application client to remote EJB session beans running on the server.

application domain A domain containing less privileged (non-system domain) classes in a protection domain.

application framework A set of classes, libraries and other software that provides a structure to help construct systems for standard types of application – for example, Apache Struts or JavaServer Faces are web application frameworks.

application gateway A kind of firewall providing a proxy service.

application object (CORBA) A user-defined object in a CORBA application.

application portability The ability to adapt an application to a different platform. The easier the adaptation is, the more portable the application is said to be.

application programming interface (API) An interface provided by a computer system or program library. Services specified in a computer system/library can be made available to, and called on by, computer programs. The software that provides the functionality described by an API is said to be an implementation of the API.

application server A server dedicated to running a number of applications.

application service provider (ASP) An organisation offering access to application software run on centralised servers and accessed over the internet using a browser or specialised client software.

application software Software designed to carry out a function for the user, e.g. a payroll application or a word processing program (as opposed to systems software like the operating system or middleware).

application-managed entity manager An entity manager that has to be explicitly created and closed by the application.

asymmetric cryptography A code in which the encoding key is different from the decoding key, but is related to it in a non-trivial way (an example of this is **public key cryptography**).

asymmetric replication See **master–slave ownership**.

asynchronous message passing A form of communication in which the sending and receiving processes do not have to be at their synchronisation points at the same time. This form of message passing requires a buffer in order to store messages in the case of the receiving process not being ready to receive the message.

asynchronous replication A strategy for replication of data, in which a replicated part is updated some time after the source database is updated.

at-most-once A semantics of invocation in which the invocation will block and will either successfully receive a result or fail due to an exception, resulting in a synchronous communication.

atomic action An indivisible action – that is, either the action happens completely, or it does not happen at all. See also **fine-grained** and **coarse-grained atomic actions**.

atomic clock An extremely accurate timekeeping device, based on the resonant frequency of atoms, usually of the element caesium, and on which official definitions for time units are now based.

atomic objects Also known as **fully synchronised objects**, atomic objects are objects which have all their methods declared as *synchronized*. The data is fully encapsulated and there is no public access to the data fields of the object apart from through such *synchronized* methods.

attribute (HTML) A named item of additional data in an HTML tag. It is usually paired with a value, such as `method="POST"` or `type="HIDDEN"`, where `method` and `type` are attributes of the `<form>` tag and the `<INPUT>` tag respectively.

attribute (of a scope object) A data item associated with a scope object in a servlet or JSP page.

attribute (XML) An attribute describes the properties of an element.

auction sniping A practice where potential bidders in a time-limited online auction access the auction for an item just before it closes and submit a bid that is higher than the current highest bid in the hope of securing the item before anyone else can raise their bid.

audit log A record of the activity on a system that can be used to determine what users have done and to assist recovery in the case of an attack.

auditing Capturing a secure record of security-related events for non-repudiation, recovery and proof of effectiveness.

authentication The process of determining a user's or software's identity.

authorisation The process of determining what code is allowed to do, according to how it is signed, what code base it comes from or what caller it is associated with (for example, what user invoked it).

availability The property of a system that can be used when it is needed by an authorised user or application.

B

bandwidth The data transfer capacity of a communications link, usually measured for digital communications in units of bits per second.

batch process A process that does not require user interaction.

before image In logging, the name given to the value of a variable before an update is made.

bidirectional relation This kind of relation can be navigated in either direction – both tables/classes in the relation point to each other.

binary semaphore A semaphore used to guard exclusive access to a shared resource.

binding The association between a name (that must be unique within a certain context) and a resource (file, service, IP address, etc.). Name services store a set of bindings and allow users to create, change and remove bindings.

BitTorrent protocol A protocol for file sharing among peers (in peer-to-peer systems), where a large data file is divided into smaller parts that are distributed among the peers. The peers cooperate by sharing the parts they have received with other peers, such that eventually all peers have all the parts of the original file.

block cipher A cipher in which encryption takes place on blocks of plaintext.

blocked process A process that is unable to proceed because it is waiting for some event to occur.

blocking When invoking methods, or calling procedures, a process is said to be blocked for the duration of the method call.

blocking semaphore A semaphore used for the synchronisation of cooperating processes.

broadcasting Sending a message to all hosts on a network; normally limited to a local network.

brute-force attack An attack in which possible keys are attempted until a correct one is found.

buffer An area of memory used for communication between processes and devices, allowing the processes or devices to operate at different speeds.

buffer overflow The error that occurs when a buffer is assigned more data than it should hold.

business interface An interface with the session bean methods that a client may call.

business logic Refers to the logic that embodies business rules rather than the view/presentation or storage of data. Typically, this would be the calculations and processes appropriate for the business.

business tier In Java EE systems, the part of the software running on the server that deals with business logic and provides the interface to the enterprise information system (database) tier.

busy-waiting A technique whereby a process repeatedly checks to see if a condition is true such as waiting for keyboard input or waiting for a lock to become available. See also **spin lock**.

bytecode The 'machine code' of the Java Virtual Machine, i.e. the result of compiling a Java program.

bytecode verifier The part of the Java Virtual Machine that checks whether the bytecode about to be executed conforms to various requirements.

C

cache An area of high-speed memory local to the CPU holding recently accessed data.

cache manager (in a transaction processing system) The part of the transaction processing system that organises the objects in main memory and interfaces with the persistent storage.

caching A technique of temporarily storing frequently accessed data in a special area to reduce the time required to gain access to the data.

cascade of aborts The situation where a transaction has had to be aborted, and this in turn leads to many other transactions having to abort.

causality The relationship between an event (a cause) and its effect – for our purposes we require a notion of time in a distributed system so that the causes occur at an earlier time than their effects

cell (of a transceiver) The area that a transceiver covers.

central processing unit (CPU) See **processor**.

centralised computer system A computer system where all, or almost all, of the processing is carried out at one central location.

certifying authority An organisation that issues a certificate associating a user with a cryptographic key.

challenge A method of authorisation in which the user is presented with a question or other task whose successful completion helps to determine their identity.

checksum A number used to check the integrity of a message.

cipher A code, a means of encrypting plaintext.

ciphertext The output of a cipher.

circular buffer A buffer (data storage area) which is logically said to be circular – when a process gets to the end of the buffer, it can start again at the beginning.

class loader A part of the Java Virtual Machine that loads classes at runtime into the Java Virtual Machine's memory. Java developers can implement their own specialised class loaders.

click fraud The fraudulent generation of clicks in order to profit from cost-per-click web pages.

client (CORBA) A CORBA object acting in the role of a client in some operation.

client polling An automatic process that causes the browser to reload a web page at regular intervals by including special instructions in the HTML header of the web page.

client-server A software architecture style in which asymmetric roles are assigned to two collaborating processes. One process, the server, plays the role of a service provider which waits passively for the arrival of a request. The other, the client, issues specific requests to the server and awaits its response.

clock skew The difference in clock readings between any two hosts in a distributed system.

coarse-grained atomic action An atomic action implemented by a sequence of fine-grained atomic actions.

code base The location of code in terms of its URL.

code on demand On a request from the client, code is transferred from the server and executed on the client, using the client's resources.

collision A collision occurs in a hash function when two or more inputs generate the same output.

commit protocol A protocol to commit transactions. See **two-phase commit (2PC) protocol**.

Committed (transaction state) The state a transaction is in after the transaction has completed successfully and all the updates to the data have been made permanent (persistent).

Common Data Representation (CDR) A set of data types that may be used to represent types in various implementation languages in CORBA interfaces.

Common Gateway Interface (CGI) A standard for interfacing external applications with information servers, such as HTTP or web servers. CGI programs are stored on the web server and are run by that server in response to particular HTTP requests.

Common Intermediate Language (CIL) In .NET, the common intermediate language to which all languages are compiled – equivalent to bytecode in Java.

Common Language Infrastructure (CLI) In .NET, the collection of technologies allowing various implementation languages to run on different platforms.

Common Language Runtime (CLR) The .NET environment that performs the task of just-in-time compilation of CIL to a platform-dependent code, and is also responsible for the lifecycle of .NET processes, including garbage collection.

Common Object Request Broker Architecture (CORBA) The part of the OMG specification commonly known as CORBA, which specifies the properties an ORB must have to be CORBA compliant.

common service In CORBA, common services are generic services that are useful across applications and domains, including support for printing, mobile agents and internationalisation.

Common Type System (CTS) In .NET, the CTS defines mappings to and from supported implementation languages and the Microsoft Intermediate Language (MSIL).

compilation The process by which source code is translated into machine code in one operation.

component model A set of standards describing the characteristics of a type of component, governing the implementation, documentation and deployment of components. Examples of such models are the Java EE framework, EJB and CORBA.

computer bureau The clients of a computer bureau pay for the computer time needed to run their business software on the bureau's computers – a similar approach to current application service providers, however this idea predates the internet so data transfer was via magnetic disks or tapes.

concurrent system A system which may have a number of activities active at the same time and allows all of its active activities to make progress.

condition synchronisation A mechanism for ensuring that a process is blocked if some condition is not fulfilled (for example, a producer process requires a buffer to be *not* full for it to be able to proceed).

condition variables Used in the operations of a monitor to provide conditional synchronisation.

conditionally thread-safe class A class that requires some external synchronisation to achieve thread safety. For example, individual accesses may not need external synchronisation, but a sequence of them might.

confidentiality The property of a system that ensures that data is visible only to authorised users.

conflicting operations Two operations *A* and *B* are said to be conflicting if, starting from some initial state, the result of executing *A* then *B* is not the same as the result of executing *B* then *A*. Also known as non-commutative operations.

consistent state A sensible state for a system as a whole (or the individual objects within the system) to be in, such that it conforms to a given specification. It usually refers to the values that are held by the variables in the system and whether these are within the range as set out by the specification.

container An environment used to deploy a component, for example a Java EE or CORBA component and providing services to that component such as support for security and transactions.

container-managed entity manager An entity manager that is implicitly created and closed by the Java EE infrastructure. By default, the entity manager automatically creates and commits transactions on method calls and returns.

contention A situation in which multiple threads are attempting to obtain the same resource.

context A record of the condition of a process at a point in time, that is, the internal state of the process (not to be confused with process state, which is a categorisation of the process at a higher level).

context awareness The ability for hardware to detect the current environment and for software to adapt to it.

context switch The saving of the context of one process and the loading of the context of another.

context switching The act of performing a context switch.

cookie An item of text that a web server may send to a client web browser as part of its response to an HTTP request. The web browser stores the cookie text, usually in a small disk file. This has many uses, such as making it easier for users to log in to a site they have visited previously or helping to maintain session information.

cooperative multitasking Multitasking that is dependent on processes voluntarily ceding way to each other from time to time.

cooperative multithreading Multithreading that is dependent on threads voluntarily ceding way to each other from time to time.

Coordinated Universal Time (UTC) The international standard time based on atomic clocks.

coordinator In the two-phase commit protocol for distributed transactions, the process that oversees the activities of other participating processes, each of which is dealing with only part of a transaction.

CORBA compliant In CORBA, software implementing the CORBA specification is said to be CORBA compliant.

CORBA services Services that are often needed in distributed applications, such as support for naming, transactions and security. See **service**.

countdown latch A flexible thread-coordination mechanism that makes one group of threads wait for each of *N* other threads to reach a certain point in their execution, with *N* being the initial value of the latch's counter.

counting semaphore A semaphore used to control access to shared resources – may be used for any number of resources.

crash resilience The extent to which an application is supported in recovering from system crashes.

credentials Security information associated with a subject, such as keys or passwords.

critical region A sequence of instructions that access shared data.

cross-compiler A compiler that will allow the creation of compiled code for a platform other than the one on which the development is undertaken.

cryptanalysis The science of decoding ciphers.

cryptographic hash function A hash function with desirable properties for applications in security.

cryptographic service A cryptographic facility provided by implementers of services for Java's cryptography API, such as ciphers, message digests and digital signing.

cryptography The science of ciphering.

custom action A user-defined JSP tag in XML format that specifies some sort of action to be carried out by the JSP page – it actually invokes some Java code from a custom tag library.

custom tag See **custom action**.

D

daemon thread A thread that runs without interaction and which the JVM does not have to wait for when exiting.

data partitioning Dividing the data resources of a system and storing them on several servers which are distributed. Each server is responsible for maintaining its own partition.

data replication Duplication of data and the storing of data on one or more servers, (possibly distributed), such that multiple backup copies exist.

data width The amount of data that a device can handle at a time.

database In computing, a structured collection of records or data that is stored in a computer so that a program can consult it to answer queries.

database driver The implementation details of a particular database are handled by the database driver for that database.

database management system (DBMS) The computer program used to manage and query a database.

database metadata Literally, data about data. For example, data about the vendor of the database system and data about the way the database records are stored.

data-oriented model A software model in which the focus is on data, and where actions are triggered as a result of data being exchanged. See also **action-oriented model**.

deadlock In a concurrent system, a situation in which two or more processes or devices are prevented, for all time, from continuing because each is waiting for one of the other processes, or for a resource that will never become free.

deciphering Converting ciphertext to plaintext.

declarative security Using an XML deployment descriptor file to delegate authorisation checks to a container.

decryption See **deciphering**.

dependency injection A technique whereby an object automatically obtains a reference to a resource it depends on. An example is the dependency injection of entity managers into session beans by the EJB container.

deployment The placing or installing of software on a target system so that it is ready to run. This process typically involves some configuration steps to tailor the software to the specific organisation, users, or application domain – for example, to specify location-specific information, such as a list of local users permitted to access the software, and the name of the local database.

deployment descriptor An XML file (automatically generated by IDEs) that defines the internal structure of each Java EE module and allows the setting of a number of attributes of the module, such as transaction behaviour, security settings, etc.

device controller A piece of hardware that can interact with a device independently of a system processor.

device driver A piece of software used to communicate with a hardware device.

digital certificate An electronic means of associating a user with a key.

digital signature An electronic means of signing a communication.

digitally signed Having an electronic signature.

directed acyclic graph (DAG) In graph theory, a graph with nodes and edges, where each edge has a definite direction (going from one node to another). The graph has no cycles, meaning that there is no path from a node that leads back to itself.

directionality (of a relation) The attribute of a relation that states whether it is unidirectional or bidirectional.

directive tag A JSP element such as `page`, `include` or `taglib` that provides information to the JSP container about the page, and affects how the JSP compiler generates the servlet that implements the JSP page.

directory service An extension to a name service that allows resources to have attributes and to find resources that match given attribute values. LDAP is a widely used directory service protocol.

dirty read A dirty read occurs if a transaction reads values that are written by another transaction that has not committed yet.

discovery service A component whose job it is to store details about services that are available on a network and respond to client requests for information about those services.

dispatcher See **short-term scheduler**.

distributed object paradigm An extension of the object-oriented paradigm to a distributed setting, in which local objects and remote objects are treated in the same way and have the same capabilities.

distributed system A system containing a number of distinct components at different locations, where each component is, in some sense, a computer system itself and the components are linked by a network. The system may demonstrate transparency to varying degrees.

distribution transparent The requirement that users can remain unaware of some of the complexities of operating in a distributed setting.

divide-and-conquer strategy A problem-solving strategy that breaks down a problem into subproblems of the same or similar type, continuing the process until the obtained subproblems are simple enough to be solved directly.

Document Object Model (DOM) A platform-independent XML validation standard defining how to read and manipulate an HTML or XML document as an object representing a tree data structure, and providing random access to XML elements.

document type definition (DTD) A DTD describes valid element types for a particular type of XML document (having a particular root element) and the relationships of the elements to each other.

Domain Name Service (DNS) An example of a name service: it looks up internet domain names and returns the corresponding IP addresses.

domain A set of classes with the same permissions.

domain service CORBA services that are needed within particular application domains, such as medical or financial systems.

drift The absolute deviation of a clock from perfect time (in seconds or a fraction of a second).

drift rate The clock deviation per unit of time (usually per second), such as typical computer hardware clocks' drift rate of between 10^{-5} and 10^{-6} seconds per second.

dumb terminal A computer terminal with very little local processing power – simply a screen and keyboard together with some simple communications links.

Dynamic Invocation Interface In CORBA, this interface allows a client to bypass a stub and interact directly with an ORB's services.

Dynamic Skeleton Interface In CORBA, this interface allows a server to dynamically create a new interface.

E

element (XML) A semantic markup of text, approximately equivalent to a type.

embedded system A computer system that is physically part of a larger system (often a real-time system) and may monitor or control the system of which it forms a part.

end-to-end encryption An approach to network security in which the sending and receiving nodes of a message implement encryption and decryption using a shared key.

Enterprise JavaBeans (EJB) The component model used by Java EE. There are three kinds of component: session beans and message-driven beans encapsulate business logic, servicing requests from clients; entities encapsulate the business data.

Enterprise JavaBeans (EJB) container The runtime infrastructure, usually part of a web server, that supports execution of EJBs.

enterprise A large organisation, such as an industrial company, a bank or a government department.

enterprise software Software that runs on enterprise systems.

enterprise system A computing system suitable for the requirements of an enterprise.

entity An instance of an **entity class** (see below), that is, an object that corresponds to persistent data in a relational database.

entity class Part of the generic Java Persistence API, which allows Java programs to interact easily with relational databases. An entity class usually just holds data and may provide business logic, although the latter is usually delegated to session beans. Entity classes must be annotated with `@Entity`.

entity manager A runtime object that manages the lifecycle of entities, allowing them to enter or leave the persistence context, to be stored in or removed from the database.

entry protocol The code that must be executed by a process prior to entering its critical region – it is designed to prevent the process from entering its critical region if another process is already in its associated critical region. An entry protocol together with its associated exit protocol should ensure mutual exclusion.

event/notification model A model in which an object can generate certain events, and when such an event occurs, it publishes a notification. The notification can be picked up by the objects that subscribed to this event.

event-dispatching thread A Java thread that can be used to respond to events in GUIs.

exclusive lock See **write lock**.

exit protocol The code that a process must execute immediately on completion of its critical region to ensure that other waiting processes may now enter their associated critical regions. Together, entry protocols and exit protocols are designed to ensure mutual exclusion.

exponential Growing according to an exponential relationship; that is, according to a power of a number.

export (a remote object) To make the remote object available so that it can accept incoming calls from clients.

extended persistence context A persistence context that has the same lifetime as an entity manager. This is the only possible setting for application-managed entity managers.

eXtensible Markup Language (XML) A form of semantic markup of text similar to HTML.

external synchronisation (time) Keeping a clock accurate by regular comparisons with a more accurate external time source, such as a close approximation to UTC.

external synchronisation (threads) Synchronisation done by the callers of methods as opposed to internally by methods themselves.

externally visible effect Any communication that an operation may have with another process, user or storage device.

F

fabrication (security) The construction of false or unauthorised information.

factory method A method that creates objects. Typically, a factory method is a static method that hides the concrete class of the returned object: the client knows only which interface the object implements.

Failed (transaction state) The state a transaction is in when the discovery is made that normal execution cannot proceed.

fail-fast operation An operation that fails as soon as it detects some violation to consistency invariants. For example, iterators over pre-Java 1.5 collections throw an exception as soon as they detect that another thread is modifying the collection.

failure atomicity A form of atomicity, also known as all-or-nothing atomicity, which is one of the properties of a transaction. It refers to the fact that if there is some kind of failure which causes a part of the transaction not to be completed, then the entire transaction should be rolled back.

failure transparency Another term for crash resilience, referring to a system's ability to create the illusion that a failure never happened.

fairness Fairness is concerned with guaranteeing that a process will be given the chance to proceed, regardless of how other processes behave.

fat client See **thick client**.

fat client Software that runs on the client machine of a client–server system to provide a considerable degree of functionality, such as a complex user interface and significant client-side processing.

fault tolerance The ability of a system to respond gracefully to an unexpected hardware or software failure. By ‘gracefully’ we mean that if a system’s operating quality decreases at all due to a failure, the decrease is proportional to the severity of the failure. This is in contrast to a naively designed system in which even a small failure can cause total breakdown.

fault-tolerant system A system that can continue normally (or perhaps with reduced functionality) after a hardware or software error has occurred.

faults (interrupts) A class of interrupt caused by hardware failure.

fetch–execute cycle The cycle by which the next instruction to be executed is retrieved and completed by a CPU.

fine-grained atomic action An atomic action implemented directly by an indivisible machine instruction.

firewall A technology to filter internet traffic between a local and external network.

foreign key A column that uniquely refers to a record in another table.

fragmentation Slicing a database into different fragments, such that each part is stored in different physical areas of storage.

fully synchronised objects See **atomic objects**.

G

garbage collection A runtime technique to free the memory occupied by objects that are no longer used. The Java Virtual Machine specification does not impose any specific garbage collection algorithm.

geotagged information Data that has been associated to a physical location, e.g. the address of a restaurant is geotagged with the latitude and longitude of its location.

Global Positioning System (GPS) A satellite system that sends continuous radio signals, thereby enabling a receiver on the Earth to compute its latitude and longitude.

global time A common value of the current time in a distributed system, in the sense that all hosts have perfectly synchronised clocks (an ideal that cannot be achieved).

global transaction A transaction that accesses and updates data in several databases (contrast with **local transaction**).

granularity of a lock A design consideration for algorithms based on locks, regarding the size of the object that needs to be locked.

green thread A thread that is not visible to the operating system and has to be created, scheduled and terminated by the Java Virtual Machine.

grid computing A way of harnessing the power of many computers distributed across the internet to solve problems requiring great processing power.

group (security) A collection of authenticated users.

guarded suspension A method invocation is suspended until a condition (i.e. guard) holds. Equivalent to **condition synchronisation**.

H

hacker An unauthorised user of a system.

handshake An exchange of information to establish parameters for communication.

hash function A one-way function that maps an input to a small output.

heap A dynamically allocated area of memory used to store objects in the JVM.

heterogeneity Programs running on different computers with different operating systems, data representations and programming languages.

heterogeneous middleware Middleware in which no assumptions are made about a shared language or whether all parties are running on the same platform. CORBA is a heterogeneous form of middleware.

heterogeneous system A system implemented using more than one kind of computer, operating system or communication protocol and particularly using more than one programming language.

hidden form field A parameter in a web form defined using an `<INPUT>` tag with the attribute value `type="HIDDEN"`, which means it is not displayed by the browser.

hierarchy of trust The relationship in which one or more parties place trust in other parties to perform authentication.

homogeneous middleware Middleware in which all parties are assumed to be implemented in the same language and running on the same platform. For example, **Remote Method Invocation** is a homogeneous form of middleware, in that it assumes that the whole system runs within the Java Virtual Machine environment and that one single programming language (Java) is used throughout.

homograph attack An attack using social engineering in which a name similar to another is used to try to deceive a user.

host An individual component of a distributed system that has some processing capability.

HTTP header The first part of an HTTP request or response preceding the body of the request or response. A response header includes information such as the number of bytes and the type of content returned, while the body typically contains web page content or other requested resource.

HTTPS A secure HTTP protocol that uses encryption to protect information sent across the Web.

I

I/O interrupts The class of interrupts concerned with input and output events.

idempotent operation An operation that can be safely repeated because execution of the operation should leave the system in the same state – for example, a database query.

identification See **authentication**.

IDL compiler A compiler for an interface definition language (IDL).

immutable class A thread-safe class whose instances never change once they have been created.

implementation repository In CORBA, a database used by a server to register information about servant objects and enabling a client to access servant operations.

implicit object An object in a component such as a JSP page or a servlet that is already available for access and does not need to be explicitly declared, e.g. a request object.

inconsistent state The state a system as a whole, or the individual objects within it, may end up in as a result of a violation of the specification for that system.

indirection The general technique of using an intermediate layer to facilitate communication between two parts of a system.

integrity The property of information that it has not been altered by an attacker or otherwise corrupted.

interception Access of information by an unauthorised user.

interface (component/system) Used to refer to the commands or operations (methods), including their syntax and semantics, that a software system or component provides to the clients of the system. Through the interface, the software entity presents methods for external communication, and the interface allows the entity to be internally modified without affecting the way outside entities interact with it.

interface (CORBA) The general term for descriptions of components' message-passing ability in terms of their operation names and parameters.

interface definition language (IDL) An IDL describes the message-passing ability of components in terms of operation names and parameters and provides a layer of indirection above the component.

interface repository A database of interface information in CORBA, similar to the registry in Remote Method Invocation (RMI).

intermediate state The state that a system is in while it is in the middle of executing an operation. In transaction processing it is desirable that other concurrent transactions avoid seeing the intermediate state of a given transaction.

internal synchronisation (time) A process where a host compares its clock value with the clocks of other hosts in the same system so that all hosts can ensure they have approximately the same clock time, although this time may differ significantly from the correct (UTC) time.

internal synchronisation (threads) Synchronisation done by methods themselves as opposed to externally by callers of methods.

Internet Inter-ORB Protocol (IIOP) A protocol for use in inter-ORB communication in CORBA applications, and providing support for distributed communication.

Internet Protocol (IP) address The numeric address of a device on the internet.

interoperability The property of being usable by other systems, particularly in a heterogeneous system context.

interpreted (programming language) A programming language where source code is translated to machine code in stages rather than all at once.

interpreter A program that executes another program.

interrupt A mechanism by which a device or software can signal that an event has occurred.

interrupt disabling A technique to allow a currently executing process to complete its execution without interruption.

interrupt handler A software routine responsible for managing interrupts.

interrupt rate limiting A technique employed by hardware to prevent a flood of interrupts being created.

interrupt storm The condition under which more interrupts are generated than can be dealt with.

interruption (security) An attack preventing availability of a resource.

inverse side In a bidirectional relation between entity classes, the class that is not the owning side.

J

Java Naming and Directory Interface (JNDI) An API for accessing a variety of naming and directory services from within Java programs in a vendor-independent way.

Java Plug-in Downloadable software that can be 'plugged in' to most browsers to provide a standard and up-to-date JVM and runtime environment enabling them to run applets using current versions of Java.

Java Runtime Environment (JRE) An implementation of the Java Virtual Machine specification, providing an environment supporting the loading and running of Java programs and support from library code.

Java Transaction API (JTA) A standard Java interface for demarcating transactions.

Java Virtual Machine (JVM) A virtual machine defined mainly with the execution of Java programs in mind. The Java Virtual Machine is a specification, and there are many implementations of it, optimised for different platforms or Java editions.

JavaBean An object of a Java class with the following characteristics: a zero-argument constructor, getter methods for every readable property, setter methods for every writeable property, must be serialisable. JavaBeans are useful in web applications for storing, or transferring, data between servlets (and JSP pages).

JavaServer Faces An MVC application framework for use in constructing web applications.

JavaServer Pages (JSP) A technology that facilitates creation of web pages with **JavaServer Pages pages**.

JavaServer Pages page (JSP page) A web page with both static and dynamic components that can be used to dynamically generate a web page in response to an HTTP request. These JSP pages are complementary to servlets and may be more convenient for web designers as they can be created without Java programming. See also **JavaServer Pages (JSP)**.

JDBC An API for the Java programming language that defines how a client may access a database.

JSP document A JSP page which uses only XML-style tags, hence allowing the JSP page to be manipulated and validated by XML tools.

JSP expression A JSP element containing a Java expression that produces a value; this value will be displayed as part of the output (different from the **JSP expression language** since a JSP expression uses Java syntax directly).

JSP expression language A language for use in JSP to write expressions that access JavaBeans components and a number of implicit objects – an alternative to using Java programming fragments.

JSP Standard Tag Library (JSTL) A JSP tag library forming a standard part of the Java EE that contains a number of commonly used sub-libraries (such as the core library, for flow control, and libraries containing tags facilitating XML processing, internationalisation and SQL database access).

JSP tag An element in a JSP page, enclosed in angle brackets, that is part of specifying the dynamic content of the page – the three main categories of JSP tag are directive, scripting and action.

JSP tag library A collection of Java code that carries out the actions associated with JSP tags, especially custom tags, hence relieving JSP developers of much of the need for Java programming in the JSP pages. The most commonly used tag library is the JSTL.

just-in-time (JIT) compilation A technique to compile a sequence of virtual machine instructions into machine code as the sequence is being executed.

K

key (cryptography) A value used to encrypt plaintext or decipher ciphertext.

key space The set of keys associated with a cipher.

key stream The generation of changing keys for the encryption of data in a stream cipher.

L

latency In general, the delay between the time an activity is initiated and the time its effects are noticed. In the concrete case of networks, it is the time taken for messages to be transmitted from one host to another.

layer A layer of software in which implementation details are hidden and where the functionality of the layer can be invoked through a well-defined interface.

leeching A term used to describe the behaviour of those peers in a peer-to-peer system that cause a drain on the overall system – they wish to benefit only from the system and do not participate in helping other peers.

legacy system A software system that is still required because it meets some useful purpose for an organisation. It is not considered to be replaceable and so is maintained, usually on a restricted budget.

lifecycle method One of several methods that are defined for all servlets to deal with initialisation, handling requests or final clean-up.

lightweight process See **thread**.

linear Growing according to a linear relationship, that is, by a constant factor for each fixed change in input.

link encryption An approach to network security in which each node implements encryption and decryption, each communicating pair using a different shared key.

linking The stage at which classes are incorporated into the runtime state of a virtual machine.

livelock A situation where a process is continually executing an operation without getting nearer to a condition becoming true.

liveness This property asserts that a program must make some form of progress, and not come to a halt or run indefinitely without achieving its goal. See also **safety**.

load balancing Balancing the load (the amount of work) in order to avoid having one component doing all the work, while other components remain idle.

loading (Java Virtual Machine) The stage at which classes are located and checked.

local area network (LAN) A computer network where all the hosts are physically close together, often in the same building, and normally entirely controlled by one organisation or department (in contrast to a **WAN**).

local clock The clock giving a time value in a particular host in a distributed system – typically local clocks show different values from each other.

local object In distributed communication, an object whose methods can be invoked only by a local process.

local process A process that runs on the machine on which the objects it is dealing with reside.

local transaction A transaction that accesses and updates data in a single local database (contrast with **global transaction**).

location transparency The ability to manipulate remote objects in the same way as local ones.

location-aware system A mobile system that can take advantage of satellite location services (such as GPS) to determine its location and hence adapt its behaviour depending on its physical location.

lock A file or object is said to be locked when processes are prevented from accessing it (usually because another client/process is accessing it). In Java, each object has a lock.

log The place in which logging information is recorded. The log is stored in stable storage.

logging The process of recording the new value and the old value and the name of a transaction so that the persistent store can be rolled back to the previous state. A log will also contain information indicating the start of a new transaction, the number of update records and the fact that the transaction has committed. Log information must be written before the data in permanent store is changed.

logical mobility The ability of a software component to change host.

logical time A system of recording time values so as to ensure that the order of events is consistent with causality – i.e. that the cause of an event (such as sending a message) occurs at an earlier logical time than the event itself (the arrival of the message). The time values are not normally related to the actual time shown on local or external clocks.

long-term scheduler A process scheduler that controls the admission of new processes to a system to achieve a desirable mix of runnable processes.

lookup service The part of a discovery service used by clients to ask for information about registered services. The discovery service can then enable clients to choose between various services, or it can choose a service for them.

loose coupling A loosely coupled component is insulated from changes in another component due to the presence of a layer of indirection between them.

lost update problem The situation arising when an update made by one process may be overwritten by another process (and hence lost). This occurs if uncontrolled access to a shared resource has been allowed.

M

machine code Platform-dependent, CPU-specific, low-level code.

many-to-many relation A relationship between two tables or entity classes, in which each record/entity of each table/class is related to zero or more records/entities of the other table/class. Usually, a many-to-many relation can be represented by two one-to-many relations with a third table/class.

many-to-one relation The inverse of a one-to-many relation.

mapping Rules for translating from a language to an interface definition language, including specification of the relationships between types in communicating idioms.

marshalling The process of taking a collection of data items and assembling them into a form (such as a byte stream) suitable for transmission in a message.

master–slave model A model of communication in which one device or process (the master) has unidirectional control over other devices or processes (the slaves). For example, in a client–server system, the client is the master and the server is the slave.

master–slave ownership Also referred to as asymmetric replication. A style of ownership of data in schemes for replication of data where data is owned by the master site, and only the master site is allowed to make updates to the data.

medium-term scheduler A process scheduler that manages the swapping of processes to and from secondary storage.

memory Short-term, rapidly accessed computer hardware for storing data – often, but not always, volatile (losing data if powered off), e.g. **RAM**.

memory barrier A special instruction to make local and shared memories consistent with each other.

memory leak Loss of memory areas due to mismanagement of memory.

memory model A specification of how and when updates to shared data have to be propagated across different memories and processes.

message authentication code (MAC) A code used to check the integrity (and in some cases identity) of a message.

message bean A type of EJB component that can be used to implement a message-oriented communication approach between Java EE components.

message digest A reduced size version of a message produced by a hash function.

messaging system paradigm A form of middleware in which messages are not sent directly from client to server, but go via the messaging system, which acts as the intermediary. The intermediary then looks after the message from that point onwards. Also known as message-oriented middleware (MOM) or the message queuing system. An example is the Java Message Service (JMS).

metadata Data that describes other data. A relational schema is an example of metadata, and so are annotations like `@Table`.

Microsoft Interface Definition Language (MIDL) The language used to describe the interface to a .NET object.

middle person An attacker interposed between a sender and receiver in location or time.

middle tier The part of a layered distributed system that sits between the client tier and the EIS or database tier. In Java EE it may have two component tiers, the business tier and an optional web tier.

middleware Any software acting as an intermediary between two components, for example client and server components, and particularly software that hides the heterogeneity of components by providing a common interface, thus facilitating interoperability.

mobile agent An execution unit that moves, together with some code and resources, from one site to another, where it resumes execution.

mobile code Code that is transferred from one execution unit to another, where it is executed.

mobile computing The use of a portable computer with a wireless network connection.

Mobile IP A standard internet protocol for communicating with mobile nodes.

mobile system A distributed computer system some of whose components are not fixed in their location, e.g. a mobile phone network.

mode (interface language) A modifier for a parameter in an operation definition, such as in CORBA's interfaces, describing what may legally be done with the parameter. For example, whether the parameter may be read from, or written to.

Model 1 (web tier) An approach to web application design in which components may include a mixture of control logic, presentation logic, and sometimes business logic.

Model 2 (web tier) An approach to web application design which uses MVC to separate each of the control logic, presentation logic and business logic into different components.

model–view–controller (MVC) A design pattern that structures a system using three categories of component: model, view and controller. The model is the abstraction of all the domain data in the system. The view is the visualisation of the model. The controller is the set of components that manage the communications between model and view.

modification (security) Unauthorised alteration of information.

monitors So-called Hoare monitors are data structures that provide operations which access encapsulated shared data under mutual exclusion. The concurrency mechanism implemented by the Java language, with use of a lock and synchronised methods, is said to be based on monitors.

monolithic architecture A design of computer system in which all, or almost all, the processing is carried out at one central location, e.g. a mainframe computer.

multicast A form of communication whereby the sender sends a message once onto a network where it can be read by many receivers who have previously registered to receive such messages.

multicast socket An extension of the standard Java socket concept, used to facilitate multicast communication.

multi-core A design of CPU chip with more than one processing unit (or ‘execution core’) that can improve performance by operating in parallel – the special case of two processing units is called dual-core.

multi-party operation An operation that involves access to several objects. Since a synchronised multi-party operation requires several locks (one for each object in the operation), the programmer should be aware of the potential for deadlock.

multiple readers, single writer A classic problem (and solution) in the study of mutual exclusion and condition synchronisation, in which only one writer is allowed to write at a time, but multiple readers are allowed to read.

multiplicity (of a relation) The attribute of a relation that states whether it is one-to-one, one-to-many, many-to-one or many-to-many.

multiprocessor A computer system with more than one CPU – this could mean anything from two CPUs to thousands of CPUs.

multitasking Execution of several tasks at the same time, whether by parallel or concurrent processing techniques.

multithreaded Containing more than one thread.

multithreaded operating system An operating system that supports multithreaded processes.

multithreaded servlet A servlet that is expected to handle concurrent requests by running multiple threads if necessary – this is the default behaviour for servlets.

mutual exclusion An approach that ensures only one process at a time can access a shared resource.

mutual exclusion protocol A set of rules (protocol) which, if obeyed by a number of processes that wish to access a shared resource, will enforce mutual exclusion.

N

name (or naming) service A facility that binds names to resources (web servers, printers, databases, etc.) so that clients can locate resources by name instead of having to know their exact location. The RMI registry is an example of a name service, with resources being Java objects.

namespace A hierarchical partitioning of names that provides a way of uniquely identifying any name in a context such as an XML document or software application.

namespace prefix The use of a namespace name as a qualifier for the simple name of an element or attribute, so forming an extended name.

native thread A thread visible to the operating system and scheduled by it.

navigation (of a relation) The ability to access the related record/entity of the given record/entity.

nested monitor lockout A situation in which a thread T blocks or waits while holding a lock on the monitor that would enable other threads to make T progress. Thread T has therefore locked itself inside the monitor, keeping the 'key', i.e. the monitor's lock.

node See **host**.

nonce A value used once to defeat replay attacks in cryptography algorithms.

non-pre-emptive multitasking A process scheduling policy allowing a process to run to completion before applying a selection function.

non-repeatable read A non-repeatable read occurs if a transaction reads the same object twice during execution and finds a different value the second time, although the transaction has not changed the value in the meantime.

non-repudiation The property of a system that prevents users from denying sending or receiving some information or performing some action.

normalised turnaround time The value q/s , where q is the turnaround time and s is the service time.

null process A process executed by the scheduler when no other processes are Runnable; can be used to perform housekeeping tasks or to simplify a scheduling algorithm.

O

object (CORBA) In CORBA, the general term for an object within a CORBA application, incorporating user-defined client and server objects as well as those implementing standard specifications such as those contained in common services.

object adaptor A component facilitating communication between an ORB and a CORBA servant.

Object Management Architecture (OMA) The architecture produced by the OMG, of which CORBA is a part; described as an 'open, vendor independent specification for an architecture and infrastructure that computer applications use to work together over networks'.

Object Management Group (OMG) The group responsible for the Object Management Architecture (OMA).

object request broker (ORB) Middleware that sits between the client and the objects the client wishes to invoke. The ORB is responsible for locating each object and communicating the client's request to the object. The object carries out the request and replies.

object serialisation The marshalling and unmarshalling of objects so that they can be sent as streams of bits across JVMs. A class whose instances may be serialised has to implement the `Serializable` interface.

object service (CORBA) A general description for operations provided by objects in the context of CORBA.

object type (CORBA) In CORBA, a type defined by an interface within a module, specifying message-passing behaviour for a class of objects in a CORBA application.

OMG Interface Definition Language (OMG IDL) The interface definition language specified by the Object Management Group for use in CORBA. See **interface definition language (IDL)**.

OMG Object Model A platform-independent description of CORBA objects, operations, interfaces and types.

one-time pad A cipher with perfect secrecy, invented in 1917.

one-to-many relation A relationship between two tables or entity classes, in which each record or entity of the first table/class is related to zero or more records/entities of the second table/class.

one-to-one relation A relationship between two tables or entity classes, in which each record/entity of each table/class is related to exactly one record/entity of the other table/class.

one-way function A function with the property that it is easy to obtain an output from an input, but hard to determine the input given the output.

Open Database Connectivity (ODBC) An interface to SQL-based databases, which aids the communication with a database and the access to the database metadata. The aim of ODBC is to enable access to a great number of databases, independent of the programming language of the application that seeks access to a database, and independent of the database systems and operating systems.

Open Systems Interconnection (OSI) model A layered, abstract description for the design of communication and network protocols.

operating system Software that interacts with the underlying hardware of a computer system and provides a layer or platform on which other software, such as application software, can run without needing detailed knowledge of the hardware.

operation (CORBA) A general term for the message-passing ability of components in CORBA applications, roughly equivalent to a method.

optimistic algorithm In concurrency control for transactions, an algorithm that assumes that conflicts will occur rarely, and that takes measures after the event. Example: **optimistic concurrency control**.

optimistic concurrency control A concurrency control algorithm (optimistic) that works with the assumption that there is not much concurrency contention between transactions. It allows transactions to proceed without locking or time-stamping, and only just before it commits does it perform a validation check. It has three phases: working phase, validation phase and update phase. See also the definition of each phase.

orphan A remote procedure call whose client (i.e. the node making the call) crashes after the request has been sent.

owning side An entity class that holds the foreign key in an underlying table. If both underlying tables involved in a relation have foreign keys, then either class can be the owning side.

P

packet The unit of information in network communication.

packet filtering A technique allowing only desirable packets to travel across a firewall.

paradigm A model, example or pattern, especially an outstandingly clear or typical example or archetype.

parallel bus A set of parallel wiring connections that can link closely positioned CPUs in a multiprocessor system.

parallel system A concurrent system that can really carry out a number of activities simultaneously because it has more than one processor.

parameter (in an HTTP request) A named item of data that is sent as part of the HTTP request as a name–value pair and can be extracted for processing by a servlet or JSP page.

Partially Committed (transaction state) The state a transaction is in after the final part of the transaction has taken place, but before the transaction has committed.

peer-to-peer (P2P) A distributed system architecture where each computer has similar capabilities and responsibilities – each can act as a server as well as a client.

perfect secrecy The property of a code in which the ciphertext does not reveal any information about the plaintext or code.

performance A measurement of how well a system deals with its workload, usually expressed in terms of its responsiveness to users and the throughput of work.

permissions Values that determine which actions are authorised within a system.

persistence context A set of entities that are mapped to corresponding records in the database. Entities in the persistence context are in the Managed or Removed state, while those outside the context are in the New or Detached state.

persistence unit The entity classes, and the relational database that they are associated with.

persistent connection The approach used by HTTP v1.1 in keeping the TCP connection open after a first request to allow for possible subsequent requests. This contrasts with HTTP v1.0 which requires a new TCP connection for each request.

persistent storage Storage that does not require power such as a disk drive, so that stored data such as a file or database outlives the execution of the program that created it. (cf. **volatile**)

personal area network A short-range network connecting various devices, typically for personal use, with each other; for example, a network connecting home entertainment devices, a personal computer and a mobile phone.

personal digital assistant (PDA) A small handheld computing device, usually with relatively limited resources of display, memory and storage.

pessimistic algorithm In concurrency control for transactions, an algorithm that assumes that there are likely to be many conflicts and that takes measures to avoid them. Examples: **two-phase (2PL) locking** and **time-stamp ordering (TSO)**.

phantom read A phantom read occurs when a transaction re-executes a query, returning a set of data that satisfies the condition in the query, and finds that the set of data has changed as a result of another recently committed transaction.

physical mobility The ability of a hardware component to change physical location.

physical portability The ability to easily carry and move a computing device from one physical location to another.

plaintext Communication that has not been subjected to encryption.

platform The combination of hardware and software underlying a program's execution.

pointer A memory address that may be manipulated in a variety of ways.

policy entry An entry in a policy file describing permissions assigned to a class according to its code base, signing or principals.

polling The technique of repeatedly checking for an event.

polyalphabetic substitution A code in which ciphertext is formed from groups of more than one character at a time.

port A numbered conduit into a host on the internet, which allows a client to request a particular kind of service and allows a server to listen for particular kinds of request on a specific port. For example, port 80 is normally used to access a web server.

portability In the context of mobile systems, this means physical portability. In the context of software development, it means application portability.

precedence graph A way of representing the order constraints among a collection of transactions (or other statements). The nodes of the graph represent the transactions, and there is a directed edge from node *A* to node *B* if transaction *A* must be executed before transaction *B*.

pre-emptive multitasking A process scheduling policy that may require a process to give up the CPU before it would ordinarily have done so.

pre-emptive multithreading A thread scheduling policy that may require a thread to give up the CPU before it has completed running.

prepare phase In the two-phase commit protocol for distributed transactions, the phase in which each process involved with dealing with part of the transaction has to send a message to the coordinator to indicate whether it intends to commit. See also **resolution phase**.

primary key A column that uniquely identifies a record among all others in the same table.

primary time server A host that has direct access to a standard time source (such as an atomic radio clock) and that makes this time value available to other hosts in a distributed system – in NTP, a stratum 1 host.

primordial loader The trusted class loader in the JVM which bootstraps the system, loading the main class.

principal An identity associated with a subject, such as the subject's login ID or an organisation's name.

priority queue A queue that orders elements according to their associated priority so that the next element to be removed from the queue has the highest priority.

priority A number used to indicate the importance of a process, thread or interrupt.

private key A cipher key that is kept secret and is used in conjunction with a public key in asymmetric cryptography.

privileged instructions Machine code instructions that may only be executed by a processor in supervisor mode.

probabilistic synchronisation An approach to clock synchronisation in distributed systems that guarantees the accuracy of synchronisation with a certain probability as opposed to a deterministic (100% certain) guarantee – the accuracy can be increased by increasing the number of synchronisation requests.

procedure call When a procedure is called, the code associated with the procedure is executed, and once this is finished and any necessary data has been returned, the execution of the main program continues with the next statement. This is similar to a method invocation.

process In software terms, a process is a program under execution.

process descriptor A data structure containing sufficient information about a process to allow it to be paused and restarted.

process identifier A unique value (ID) distinguishing one process from another.

process scheduler A general term for software that manages process state transitions according to a scheduling policy.

process state A high-level categorisation of a process in terms of what it may do next, for example, Runnable or Blocked.

process table A data structure used by an operating system to store information about processes.

processor The part of a computer system that carries out the instructions of a software program in order to process data.

producer–consumer model A model in which two processes work together and share data in such a way that one process assumes the role of producer (i.e. writes to the data resource) and the other assumes the role of consumer (i.e. reads from the resource).

program counter A pointer to the next instruction to be executed in a program.

program interrupts A class of interrupt generated by software due to an exception.

programmatic security Explicit authorisation checks by method invocation.

prologue The first part of an XML document, which declares its version number and may include information such as the encoding (character set).

property (of a JavaBean) This loosely corresponds to the instance variables or attributes of a JavaBean class. Usually the property is implemented directly as an instance variable but it need not be. JavaBeans must have a getter and setter method for each property.

protection domain The interaction of features enforcing security policies in Java, particularly as regards permissions allocated to classes.

protocol (for communication) The rules used in a communication, such as HTTP for the Web, or, more generally, rules governing the interaction between different components in a system.

provider (security) A supplier of cryptographic products.

proxy object An object that ‘stands in’ for another object. Calls to the proxy’s methods are redirected to the referred object.

proxy server A server that acts as an intermediary between clients and other servers.

pseudo-parallel system A concurrent system which shares processor time between a number of activities but only one activity is actually running at any given time.

public key A key used in combination with a private key that may be publicised without compromising security of communication.

public key cryptography Cryptography in which public and private key pairs are employed; an example of asymmetric cryptography.

public key infrastructure (PKI) A system for the association of keys with users and the management of key-related information.

publish/subscribe messaging model A communication model in which an object that generates events publishes the type of event it makes available for observation by other objects; and objects that want to receive notifications subscribe to events that are of interest to them.

pull technology An informal description of the request–response nature of HTTP – that information travels from the server to the client only when the client does something to ‘pull’ that information to it.

push technology An informal description of various ways to mitigate the request–response nature of HTTP by enabling the server to send (or ‘push’) information to the client, in order to keep the client up to date, without an explicit request from the client.

Q

qualified name A name formed from a prefix and a simple name in XML.

quantum A time during which a process may run under a pre-emptive scheduling policy.

query language The language that is used with a relational database in order to query the database and retrieve information.

R

race conditions Race conditions exist if the final outcome of a process is dependent on the timing or sequence of other processes (as if the processes are ‘racing’ each other).

radio-frequency identification (RFID) A radio-based technology to communicate with small devices that can store and change small amounts of data.

random access memory (RAM) Short-term, rapidly accessed hardware for storing data – normally in the form of volatile semiconductor devices (loses data if it loses power).

read lock Also referred to as a shareable lock. A lock that needs to be taken out if a process needs to read from a data resource. Several processes can hold a read lock.

realm A set of valid users and groups for an application.

real-time system A system which must guarantee to respond correctly to inputs within a specified time period.

record In a relational database, a record is a set of values for one entry into a database.

recovery manager (in a transaction processing system) The part of a transaction processing system that deals with the recovery from crashes.

redundant array of independent disks (RAID) A redundant array of independent drives (or disks), also known as a redundant array of inexpensive drives (or disks). RAID is an umbrella term for data storage schemes that divide and/or replicate data among multiple hard drives. They offer, depending on the scheme, increased data reliability and/or I/O performance.

re-entrant locking The locking mechanism used in Java. If a thread holds the lock for an object, and it invokes another `synchronized` method *for the same object*, it is allowed to continue. This means that under these circumstances no deadlock will arise.

reference A value that merely provides access to an associated object, as opposed to providing direct access to a memory location or the ability to manipulate memory addresses.

registers High-speed memory, local to the CPU, used to execute instructions.

registration service A component providing a way of registering interfaces in a repository to allow a client to perform service discovery.

registry A naming service that allows names to be bound to object stubs for later lookup. Names have to be unique within a registry. Registries usually run in their own JVMs in the server host.

relation In a relational database, a relation is also known as a table. The table has a fixed number of uniquely named columns, and each row represents a record for an entry into the database.

relational database A database model, where the data is conceptually modelled as a relation. It uses a collection of tables to represent both the data and the relationships between the data.

relational schema The structure, in terms of tables and columns, of the data in a relational database.

release phase The phase in two-phase locking in which the locks are released, and in which no more locks are acquired.

remote evaluation The transfer of code from the client to the server, where it is executed using local resources, after which a reply is sent to the client.

remote interface An interface that extends `Remote` and declares a set of remote methods.

Remote Method Invocation (RMI) A form of middleware, and the Java version of remote method calling, allowing methods in one JVM to invoke methods in another JVM. With RMI, methods can be invoked on objects whether they are local to or remote from the invoking client.

remote method A method that can be called from another JVM. Remote methods may throw a `RemoteException`.

remote object An object that implements a remote interface. The object may provide other methods that are not to be called remotely. Remote object classes usually extend `UnicastRemoteObject`.

remote procedure call (RPC) A call to a procedure located on a machine that is different from the machine from which the call is sent.

remote process A process running on a different machine from (but connected to) the one on which the object it is dealing with resides.

remote reference See **stub object**.

replay An attack in which intercepted or stored information is reused to gain advantage.

request dispatcher An object available to servlets (and JSP pages) that can be used to send an HTTP request to another servlet (or JSP) to produce some, or all, of the response.

request-response A communication approach between software components where one sends a request and the other returns a response, for example HTTP.

reserved characters Characters in a URL that have a special meaning, such as `/` or `#`, and so cannot be used as part of identifiers.

resolution phase In the two-phase commit protocol for distributed transactions, the phase in which the coordinator decides on whether or not the transaction as a whole can commit or not, based on the responses received from the processes involved in carrying out parts of the transaction.

resolving a name The process by which an object reference is associated with a particular servant object in a CORBA application; a binding of a reference to an object name.

responsiveness The ability of a system to remain acceptably fast in its response to a request for action or information (the definition of what is acceptable depends on the context). If a system has this property, we say it is responsive.

rich client See **thick client**.

role (security) A collection of identities or category of identities with shared permissions.

rolling back The act of restoring the state of a persistent store to one that existed before the start of an atomic operation that was aborted due to a system crash or exception.

root certificate A certificate granted to a certifying authority.

root element The first element in an XML document, which encloses other elements and acts as a way of identifying the type of a document.

RSS (Really Simple Syndication) A technology that allows users to sign up to receive updates from a website whenever something important has changed.

Ruby on Rails An approach to the development of web-based access to databases that uses a database specification, and scaffolding to provide the most commonly required operations.

runnable process A process that may be selected to be run; same as a ready process.

running process A process that is being executed by the CPU.

runtime environment The software environment supporting loading, linking to libraries and execution of programs.

S

safe method A method without any side effects for which users are held responsible – for example, a database query because it does not change any stored data.

safety This property asserts that nothing bad happens during the execution of a program. See also **liveness**.

salt A number used to prevent collisions in hash functions, or more generally to add variability to the processing of information in security systems.

sandbox A restrictive environment in which untrusted Java applets execute. The word is derived from an American term for a children's sandy playing area.

scaffolding The technique of providing an infrastructure delivering commonly required operations and allowing modification for special cases, so facilitating rapid development.

scalability The ability of a system to scale (to grow, or less commonly shrink) in order to support changes in demand, for example, different volumes of data or users. If a system has this property, we say that it is scalable.

scalability under contention The ability of an algorithm to maintain a high throughput under contention, i.e. to keep a high ratio of number of threads per time unit that access the same resource.

schedule (for transactions) A sequence of operations that specifies the chronological order in which the operations of concurrent transactions are executed.

scheduler (in a transaction processing system) The part of the transaction processing system that puts together serialisable schedules for the operations of transactions.

scheduling Allocation of processor (i.e. CPU) time between the various activities in a concurrent system.

scheduling policy A policy contributing to process state change at any given time and in particular determining which process is allowed to run.

scope (of an object) The extent over which an object or other item is valid or accessible. For example, items that have session scope continue to exist throughout the duration of the session in which they were created.

scope object An object associated with a particular scope in a web application, such as the JSP page, the request, the session, or the whole application. Scope objects can be used to store or transfer data between components such as JSP pages and servlets. See also **attribute**.

scope of discovery The extent to which a service discovery application will search for a service matching a request.

scripting tag An element in JSP, with several possible types, that allows inclusion of Java declarations, Java executable statements, or Java expressions in the JSP page.

scriptless JSP pages can be specified as scriptless so that no Java-style scripting tags will be accepted by the compiler, hence enforcing use of JSP EL and JSP tags.

scriptlet A fragment of Java code in a JSP page, consisting of one or more complete Java statements that will be executed when the JSP page runs.

secret key A key that must be kept secret, particularly as used in symmetric cryptography.

secure channel A communication channel between a pair of processes providing authentication, confidentiality, integrity services and time stamping.

security manager (JVM) A component of the JVM concerned with authorisation of what code may or may not do. The security manager enforces a security policy.

security policy (JVM) A collection of permissions contained in a policy object, usually determined using policy files. The policy defines what code may, or may not, do at runtime.

security through obscurity The discredited notion that secrecy of design alone can provide security.

selection function A function used by a scheduler to select which process is allowed to run.

semaphore A type of object that is used to protect a resource from concurrent access by multiple processes. It does this through the use of two operations *semWait* and *semSignal*. Internally, a semaphore has a data field that holds its value and a queue that is used to hold blocked processes.

sequential system A system in which each activity must complete execution before another activity can start.

serial schedule A schedule for the operations of concurrent transactions, in which the operations of one transaction all occur before the operations of the next transaction, such that the transactions execute in a strictly serial fashion. There is no interleaving.

serialisability (databases) If a schedule in which transactions are interleaved is equivalent to a serial schedule it is said to have the serialisability property or to be serialisable. A serialisable schedule is also said to have serial equivalence.

serialisable (databases) A schedule for a group of concurrent transactions is serialisable if, and only if, it produces the same result as if the transactions had executed in some serial order.

servant An object, under the control of a server, providing a CORBA operation to a client.

server (CORBA) A CORBA object acting in the role of a server in some operation.

ServerSocket A type of object used by a server program in a client-server application. The object has the role of waiting for a client and returning a connection socket once the connection with this client is established.

service A generic term describing the message-passing ability of components, but particularly used in service-oriented systems.

service time The time a process would run if it had sole use of the CPU.

service-oriented architecture An architecture in which the main unit of communication is a message as opposed to a method invocation. Service providers use an interface to describe their services, and can be connected together in arbitrary configurations.

servlet A Java program that runs on the server and that follows a request–response model, normally responding to HTTP requests.

session A conversation consisting of a series of linked requests and responses between the client and server on behalf of a particular user.

session bean A type of EJB that handles an interactive session between a client and the application server. Session beans are neither persistent nor shared.

session ID A number or string that uniquely identifies a particular session and any associated data.

session key A key that is used only while a session is in effect and then discarded.

shadowing Where the results of a transaction are built up in a structure that mirrors the persistent store but the persistent store is not updated until the transaction commits. Once the transaction has committed, the shadow structure replaces the persistent store.

shareable lock See **read lock**.

shared resources A resource, which may be either data (in the form of variables or files) or hardware, that a number of processes have access to.

short-term scheduler A process scheduler concerned mainly with which process is allowed to run next.

Simple API for XML (SAX) An API providing access to XML as a stream of data and allowing serial access to elements, but not random access.

simple name A name used to uniquely identify an XML document's namespace.

single anonymous condition variable Java's concurrency mechanism allows just one wait set for all the threads that have called `wait` on a given object's lock. This acts as the condition variable. This is in contrast to so-called Hoare monitors which can have multiple condition variables, and a queue for each specific condition. Therefore, we say that Java has a single condition variable, which is anonymous as it does not belong to any particular condition.

single point of failure The part of a system that can cause the whole system to go down when a failure occurs in this particular part.

single-threaded servlet A servlet that is limited to handling only one request at a time – this approach is not now recommended.

singleton A programming design pattern ensuring that only one of an object may exist at a time.

skeleton A layer between a servant and an ORB in a CORBA application.

sniping See **auction sniping**.

SOAP An XML-based protocol supporting communication between distributed systems and widely used in web services.

social engineering A collection of techniques used to manipulate people into performing actions, or divulging confidential information. The term typically applies to trickery associated with information gathering or accessing computer systems.

socket An abstraction for each of the two sides of the connection between the client and server. It is identified by a combination of port and host name, and provides the facilities (e.g. input and output streams) to enable data to be transferred between client and server. In Java these are implemented with the `Socket` class.

software architecture A software architecture is concerned with the structures of the system. These structures comprise software elements, the main properties of those elements and the relationships between them.

software component A software element that conforms to certain requirements for interoperability defined by a component object model, for example, provision of an interface through which services are provided, hiding implementation details. In particular, software components support the goal of composability to create higher functionality than they individually provide.

spin lock A lock where the thread simply waits in a loop ('spins'), repeatedly checking until the lock becomes available. This is also known as **busy-waiting** because the thread remains active but is not performing a useful task.

stable storage An abstraction which means that the results of an atomic operation are written to two independent places in a persistent store.

stack A last-in, first-out (LIFO) data structure, used to store activation records for Java threads.

standard action One of the JSP tags that form a standard part of the JSP specification and that cause some sort of action to be carried out by the JSP page (as opposed to custom actions which are user-defined via custom tag libraries).

starvation Starvation occurs when a runnable process is overlooked indefinitely by the scheduler.

stateful packet inspection A firewall technique in which the state of a system is used to determine the legality of traffic across a firewall.

stateful session bean A session bean that implements a single business service that spans multiple method calls, hence requiring state to be kept between calls.

stateless protocol A communication protocol, such as HTTP, where no state information is stored and each request or other communication is independent of any others.

stateless session bean A session bean that does not keep state between calls, because each method implements a separate service: each session consists of a single method call.

statically typed Checking typing of variables at compile time.

statistical analysis A cryptanalytical technique using patterns of frequency of letters in languages as an aid to deciphering messages.

storage Computer hardware for holding data in the long term (even after being powered down) – typically uses devices such as magnetic or optical disks that are slower but cheaper per unit of data than memory.

stream cipher A cipher whose key changes frequently to encrypt a stream of data.

strict two-phase locking (strict 2PL) A form of two-phase locking in which all locks are released on commit.

strong mobility The ability, provided by the platform or middleware, to suspend an execution unit, capture its state (including the code it is executing and the data it is manipulating) and transfer all this to a new site, where execution is resumed.

strong typing Checking for potentially troublesome conversions of types at compile time.

Structured Query Language (SQL) The most widely used and standardised query language for relational databases.

Struts framework An MVC application framework for use in constructing web applications.

stub A layer between a client and an ORB in a CORBA application.

stub object A proxy for a remote object. There is a copy of the stub object on the registry and on each client JVM that wants to use the corresponding remote object.

subject (for authentication) A grouping of related information for an entity in a security system, such as a collection of identities (principals) and credentials.

substitution A ciphering technique in which one character in a language is substituted for another.

supervisor call interrupt An interrupt used to signal that supervisor instructions are to be executed.

supervisor mode The mode a CPU enters to execute supervisor instructions.

suspended process A process that is stored on a disk; same as swapped out.

swapped out See **suspended process**.

swapping The action of moving a process from volatile to persistent storage.

swarm A group of connected peers working together in a peer-to-peer system.

symmetric cryptography Cryptography employing a private key or in which the decoding key is trivially related to the coding key.

synchronisation (processes) The need for different processes to synchronise their operations by waiting for each other. In Java, the keyword `synchronized` is used for methods, so that only one thread can execute the method at a time, forcing the other threads to wait.

synchronisation (time) To cause clocks to indicate the same time or time values as closely as possible.

synchronous message passing A form of communication in which the sender waits (i.e. it is blocked) until it receives a response from the receiver. See also **asynchronous message passing**.

synchronous replication A strategy for replication of data, where all copies are updated at the same time.

system In the context of security, refers to the hardware, platform, application software and users' interactions.

system bus A hardware channel for data transmission between device controllers and CPU.

system domain The collection of classes within a protection domain that are trusted.

T

tag An element of a markup language, such as HTML or XML, or in JSP, with a variety of meanings and uses. In JSP, tags are mainly used to include fragments of Java code, or equivalent items such as JSTL and JSP EL, in order to generate the dynamic parts of web pages.

tag handler The Java code that is executed when a JSP page containing its associated custom tag is run – this code is stored in a tag library.

target An entity to which a permission is applied in a policy file.

task A sequence of instructions treated as an element of work. A task may be represented by a method in a Java program, for example, but the term is more widely applied and can refer to elements of work at a low (processor) or high (application) level.

TCP/IP TCP (Transmission Control Protocol) and IP (Internet Protocol) are the most commonly used specifications of the detailed format of messages exchanged between internet hosts.

terminal A device for accessing a centralised computer system via local cabling or, if more remote, via the telephone system. Some terminals have limited processing ability, often to aid in graphical display, but the bulk of processing in the system is carried out on the central processor.

test-and-set (TAS) A special atomic operation that tests and sets a Boolean, provided by the operating system as an indivisible machine instruction.

thick client Complex client software that carries out significant processing on the client machine in a client–server system.

thin client In a client–server system, a simple client with very limited functionality, mainly limited to display and communication – for example, a web browser.

thread A lightweight process; a thread of execution.

thread of execution A path of execution within a process.

thread pool A set of threads, each one taking a task from a common queue, executing it and then fetching the next task. The main aim of the pool is to reuse threads for multiple tasks, in order to optimise the memory usage and the performance of the application.

thread safety A class is thread safe if it can be executed concurrently by multiple threads without the shared data getting into an inconsistent state.

thread safety The property of a class behaving under concurrent access in the same way as under sequential execution. There are various levels of thread safety, depending on how much external synchronisation is necessary to achieve it.

thread-compatible class A class that provides no internal synchronisation. As such, access to it requires external synchronisation within a concurrent application.

thread-hostile class A class that is not safe to use in concurrent programs. No amount of external synchronisation will make instances of such classes behave as if they were accessed sequentially.

thread-safe Shared data, or program components that access it, are thread-safe if there is no possibility of data loss or corruption when a number of concurrent threads are active.

thread-safe class A class that has mutable instances and where the effect of any concurrent access to an instance could also be obtained by some sequential execution. Thread-safe classes do not require any external synchronisation.

threat monitoring The active consideration of dangers to a system’s security.

throughput The rate at which computational work is done. This may be measured in a variety of ways, for example, by the number of processes completed per unit of time by a CPU or the number of transactions completed per second in a database system.

tier The common functionality of computer processes or components at a very high conceptual level.

tight coupling A tightly coupled component communicates directly with another in such a way that changes in one are liable to affect the other.

time stamping The inclusion of information with a message so that its time of occurrence can be determined.

timeout The expiry of a preset time interval causing a thread to stop running because it has used its allocated execution time.

timer interrupts A class of interrupt used to keep track of the passage of time; used by schedulers to pre-empt processes.

time-sharing A rather dated term for a concurrent system with a number of users that allocates time to each user, so that each has the impression that the computer is responding to just their request.

time-slicing A scheduling approach for concurrent systems in which each activity gets a fixed and equal amount of processor time before the processing switches to the next activity.

time-stamp A record of the time of an event (such as the sending of a message) that is attached to that event (such as being appended to the message in a data packet).

time-stamp ordering (TSO) An algorithm for concurrency control for transactions based on time-stamps. A time (such as the time at which the transaction started) is associated with each transaction and is recorded by every object involved with each operation invoked by the transaction. This time-stamp is used to serially order transactions.

topic In a messaging system (such as JMS), a topic is a named area of interest to which clients can subscribe, such that they receive any messages relating to this area.

topological sorting In graph theory, a topological sort of a directed acyclic graph is a linear ordering of its nodes such that each node comes before all nodes to which it has edges. Every directed acyclic graph has one or more topological sorts.

transaction A sequence of atomic actions, with the property that either the entire sequence is performed, or no part of the sequence is performed at all.

transaction isolation level The levels that allow fine-tuning of transactions, based on different reads: dirty reads, non-repeatable reads and phantom reads.

transaction manager (in a transaction processing system) The part of a transaction processing system that provides an interface between the clients and the system, and which passes on requests for transaction processing to the scheduler.

transaction processing (TP) system A layer of middleware to manage the correct and efficient execution of transactions. Also known as a transaction processing (TP) monitor.

transaction-scoped persistence context A persistence context that has the same lifetime as a transaction. This is the default setting for persistence contexts used with container-managed entity managers.

transceiver A transmitter and receiver of radio wave signals.

transfer object An object used for transferring aggregated data between clients and session beans.

Transmission Control Protocol (TCP) connection A link agreed between two internet hosts that is established by a 'handshake' (an exchange of a number of initial data packets), prior to the hosts sending actual data.

transparency The extent to which a distributed system appears to its users as a coherent system providing a range of functions (so that the user does not need to know that the system is distributed).

transposition An encryption technique where the order of information in the plaintext is obscured.

trapdoor function A one-way function which is reversible if given some secret information.

trusted applet An applet that can guarantee its origination from a known source by means of cryptography, and may therefore be permitted to carry out a wider range of actions than an untrusted applet.

tunnelling Wrapping one protocol in another protocol.

turnaround The time interval between the submission of a process and its completion.

two-phase commit (2PC) protocol A protocol for distributed transactions, where one process acts as the coordinator for several parts of a transaction. All parts of the transaction must agree to commit in order for the transaction to commit as a whole. There are two phases: the prepare phase and a resolution phase.

two-phase locking (2PL) An algorithm for concurrency control for transactions based on locking. Locks are acquired for a transaction as they are needed (acquire phase), and no lock can be released until all locks have been acquired. When locks are released, no more locks can be acquired (release phase).

two-tier model A software architecture consisting of a database tier and a client applications tier.

type safety The ability of a language to check that types are being used correctly.

U

ubiquitous computing A situation where computing devices are so small and cheap that they occur in huge numbers throughout everyday environments.

unicast A point-to-point form of communication, that is, involving one sender and one receiver at a time.

unicast communication Point-to-point communication.

unidirectional relation This kind of relation can be navigated in only one direction, namely from the table/class that holds the foreign key to the other table/class.

uniprocessor A computer system with a single central processing unit (CPU).

Universal Description, Discovery and Integration (UDDI) An XML-based standard used by a server to register WSDL descriptions of web services and by a client to discover such services. As such it allows the creation of directories for web services.

unmarshalling The process of extracting a data structure from a series of bytes.

unresponsive Unacceptably slow to respond to a request for action or information – the definition of acceptability depends on the context.

unsafe character One of the reserved characters or other characters that cannot be included in a URL without the possibility of confusion about their intended role or meaning.

untrusted applet The default situation for an applet automatically downloaded along with a web page and subject to severe restrictions in the interests of security.

update anywhere ownership A style of ownership for data in schemes for replication of data, where anyone is allowed to update. Often used in the peer-to-peer environment of open source projects.

update phase (in optimistic concurrency control) The phase in which a transaction has been validated, and can now commit.

URL encoding A means of including potentially unsafe characters in a URL by encoding them using the corresponding ASCII code, such as using %2F instead of /.

user mode The CPU mode when it is executing ordinary (not supervisor) instructions.

user thread A thread created by a user; a non-daemon thread.

UTC See **Coordinated Universal Time**.

V

valid An XML document is valid if it is well formed and conforms to a DTD.

validation phase (in optimistic concurrency control) The phase in which a transaction gets validated. If validation succeeds, the transaction can commit.

vector A route by which an attack may occur on a computer system.

virtual machine (VM) An abstract computing device implemented in software. A virtual machine defines a certain set of instructions and how code and data are organised in the machine's memory.

virtual memory External memory; persistent storage used to supplement main memory.

virtual private network (VPN) A network created by a form of tunnelling in which end-to-end encryption takes place between the two ends of the tunnel, so creating a private network of sorts.

volatile storage Storage that requires power in order to retain any data stored in it, for example RAM, cache and registers.

volatile variable A variable declared with the `volatile` modifier. For practical purposes it is as if such variables are always accessed directly in shared memory. Moreover, an access to a volatile variable creates a memory barrier.

W

wait set The set of threads that have called `wait` and are waiting to be notified. Each object in Java has one wait set.

wait-notify mechanism The Java mechanism to implement condition synchronisation. A thread calls `wait` and will be blocked as long as the condition it is waiting on is not satisfied. Another thread calling `notify` or `notifyAll` can release this thread from its blocked state.

weak mobility The ability, provided by the platform or middleware, to transfer code from one site to another.

weakly consistent operation An operation that does not guarantee consistency invariants and as such does not fail under concurrent accesses. The iterators returned by the concurrency classes introduced in Java 1.5 are weakly consistent and never throw an exception due to concurrent modification.

web container A software environment in which servlets and JSP pages run, and that provides a number of services including security, concurrency and management of the servlet lifecycle.

web form A portion of a web page that allows the user to enter data to be sent, in the form of an HTTP request with associated parameters, for processing by the server.

web of trust A non-hierarchical approach to establishing trustworthiness of digital certificates.

web service An example of the network services paradigm in which software services are provided via a Web-based application, particularly using HTTP, web pages and SOAP as communication mechanisms.

Web Services Description Language (WSDL) A language used by UDDI for the description of web service operations using XML.

web tier The part of the middle tier in a Java EE application that communicates with the client tier, typically by receiving and responding to HTTP requests using dynamically generated web pages.

- well formed** The property of an XML document of being syntactically (as opposed to semantically) correct.
- wide area network (WAN)** A computer network that spans a large geographical area such as a region, a country or the world and which typically relies on public providers of telecoms or other long-distance services for part of its communication path (in contrast to a **LAN**).
- workflow ownership** A style of ownership of data in schemes for replication of data, where the right to make updates to replicated parts moves from site to site.
- working phase (in optimistic concurrency control)** The phase in which a transaction works with a tentative version of the objects.
- wrapper method** A method that adds new functionality or properties to existing objects. The `Collections` class provides static wrapper methods to turn thread-compatible collections into conditionally thread-safe ones.
- write lock** Also referred to as an exclusive lock. A lock that needs to be acquired by a writer process. Only one process can hold a write lock at the same time.
- write-ahead log** In logging, a log must be written ahead of the actual update made to the data, in case a crash occurs just after an update is made, and before this update could be recorded in the log.

X

XML See **eXtensible Markup Language (XML)**.

XML Schema Definition (XSD) An XML-based description of an XML document, allowing for better type checking than a DTD and more rigorous constraints for XML documents to meet.